

KHELIFI ALAE

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PROFESSIONAL SUMMARY

Master's student in Artificial Intelligence at UMBB, seeking a 2026 summer internship in AI, Data Science, or Software Development. Experienced in deep reinforcement learning, computer vision, and real-time data pipelines. Passionate about applying AI to solve real-world problems.

EDUCATION

Master in Artificial Intelligence (M1) 2025 – 2026, *Present*
CS Department, UMBB

- Advanced Algorithms, Machine Learning, Computer Vision, Deep Learning, Cloud Computing, Advanced Database Systems.

3rd Year Bachelor's Degree 2022 – 2025
CS Department, UMBB

- Program focused on information systems and industrial logistics, involving programming, database design, web development, and data analysis, with a final project on an AI model (Final project graded 17/20).

High School Diploma - Mathematics Option 2022
Zemmouri El-djadida High School, Zemmouri

- Average obtained: 14.83.

PROJECTS

INDUSTRIE IA — Multi-Agent Manufacturing E-Commerce Pipeline / Reinforcement Learning for Autonomous Lunar Landing April 2026
GitHub

- Orchestration Agents:** Engineered a multi-stage industrial AI pipeline using LangGraph and FastAPI, transforming raw technical PDFs into structured manufacturing dossiers and client-ready deliverables.
- LLM OCR Integration:** Built an extraction engine (M1) using Ollama (Mistral/Qwen) and Tesseract OCR to normalize complex industrial specifications into validated JSON schemas.
- Financial Predictive Analytics:** Developed a 10-year Total Cost of Ownership (TCO) calculator integrated with World Bank APIs for inflation-adjusted forecasting and a Digital Twin module for predictive maintenance.
- Automated Content Generation:** Implemented dynamic 3D video generation using Manim and automated CAD asset creation (DXF/IFC) based on real-time extraction results.
- Infrastructure:** Containerized the entire microservice architecture using Docker Compose to ensure modularity and reproducibility.
- Technologies:** LangGraph, FastAPI, Docker, Ollama (LLMs), PostgreSQL, Manim, Pandas, World Bank API.

Biometric Iris Recognition System (Siamese-ResNet)

February 2026

GitHub

- Architecture and Optimization:** Developed a deep learning pipeline for high-precision biometric authentication using a modified ResNet-18 backbone and Siamese Networks.
- Contrastive Learning:** Implemented Contrastive Loss optimization to minimize vector distance between authentic pairs while maximizing separation for impostors, achieving a state-of-the-art Equal Error Rate (EER).
- Robust Preprocessing:** Built an automated data pipeline on Kaggle to process bilateral iris textures (left/right eye variations), ensuring model reliability across diverse imaging conditions.
- Evaluation:** Validated system performance through distribution analysis of embedding distances and real-world inference testing on physical photographs.
- Technologies:** Python, PyTorch, ResNet-18, OpenCV, Kaggle, Scikit-learn.

Real-Time E-Commerce Analytics Pipeline November 2025
GitHub

- Architected event-driven pipeline with Apache Kafka processing 10,000+ orders/hour, reducing latency from 24hr batch to <5 sec real-time through stream processing and aggregation.
- Containerized with Docker Compose (Kafka, Zookeeper, PostgreSQL, Streamlit) – Python, Pandas, Docker.

Breast Cancer Classification with Machine Learning

October 2025

GitHub

- Developed KNN model achieving 96.5% accuracy for tumor diagnosis; applied PCA reducing features 30→10 while maintaining 95%+ accuracy and 3x faster inference.
- Technologies:** Python, scikit-learn, Pandas, NumPy, Matplotlib, PCA.

Automated Exam Scheduling System December 2025
GitHub / Live

- Built constraint-based scheduler for 2,000+ students using genetic algorithm, reducing planning time from 2 weeks to 15 min – deployed on Railway, piloted by CS department Spring 2025.
- Technologies:** Python, Streamlit, MySQL, Genetic Algorithms, Railway.

RELEVANT SKILLS AND EXPERIENCE

Soft skills: Communication, Analytical and Problem Solving, Teamwork, Adaptability.

Proficient: Python, PyTorch, Machine Learning, Deep Learning, Reinforcement Learning, Git/GitHub, SQL, PostgreSQL.

Experienced: TensorFlow, OpenCV, Docker, Apache Kafka, Streamlit, MySQL, C/C++.

Familiar: JavaScript, Node.js, Flask, PowerBI, SharePoint, MongoDB, R, Java.

LANGUAGES

French: Fluent · **English:** Fluent · **Arabic:** Native language